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Adversarial training via multi-guidance and historical memory enhancement

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ABSTRACT

Deep neural networks (DNNs) are often susceptible to the influence of adversarial examples, potentially leading to severe security issues. Adversarial training stands out as one of the most effective defenses. In this paper, we empirically demonstrate that weakly robust models maintain resilience to adversarial examples generated on a fully trained robust model. Building on this observation, we introduce a novel weight ensemble training method named Multi-Guided Adversarial Training (MGAT). MGAT improves both clean and adversarial accuracy through two guiding strategies: self-guidance and expert-guidance. Self-guidance promotes diversity in decision boundaries among member models, while expert-guidance aligns them more closely with natural decision boundaries. Additionally, MGAT enhances the historical memory of model weights using an exponential moving average (EMA) with a memory factor, thereby better memorizing the weights of weakly robust models. Our experiments demonstrate that MGAT is highly effective in defending against a range of adversarial attacks. Notably, MGAT achieves an accuracy of 55.48% against AutoAttack, marking a 10.92% improvement over standard adversarial training on CIFAR-10.

1. Introduction

Due to the significant breakthroughs brought about by artificial intelligence in real-world applications, deep neural networks (DNNs) have been widely employed in critical domains. However, DNNs are often susceptible to the influence of adversarial examples [1,2], potentially leading to severe security issues. To enhance the adversarial robustness of models, numerous defense methods have been proposed, including defense distillation [3], feature squeezing [4], randomization-based methods [5], and adversarial training [6–9]. Adversarial training is currently regarded as the most effective approach for improving model robustness, both in white-box and black-box settings [8].

Traditional adversarial training focusing on the training of a single network, and improves the model robustness by data augmentation, adding regular terms, and so on. Compared to traditional adversarial training, ensemble adversarial training demonstrates better robustness enhancement. They mainly improve model robustness by combining adversarial perturbations generated by different models [8], increasing the diversity of member models [10], and ensemble each historical model on the optimization trajectory [11]. However, these ensemble adversarial training methods ignoring the interactions between individual members. It may reduce the improvement of robustness [12,13].

During the standard adversarial training [7], we experimentally found that weakly robust models at the early stages still have relatively impressive accuracies for adversarial examples generated on robust models (see Fig. 2). Therefore, when ensembling, we should consider integrating both weakly robust models at early stages and up-to-date models optimized by the optimizer. Moreover, the adjustment of decision boundaries based solely on adversarial examples may lead to overfitting of the member models [14]. Integrating these member models can potentially lead to a decrease in the clean accuracy of the ensemble model (see Fig. 1). Even when information from clean examples is incorporated using losses from traditional adversarial training, the effect on improving model robustness is not significant [11].

To address this issue, we need to seek a new ensemble adversarial training method that considering the interactions between the member models to improve the robustness. We propose a novel method

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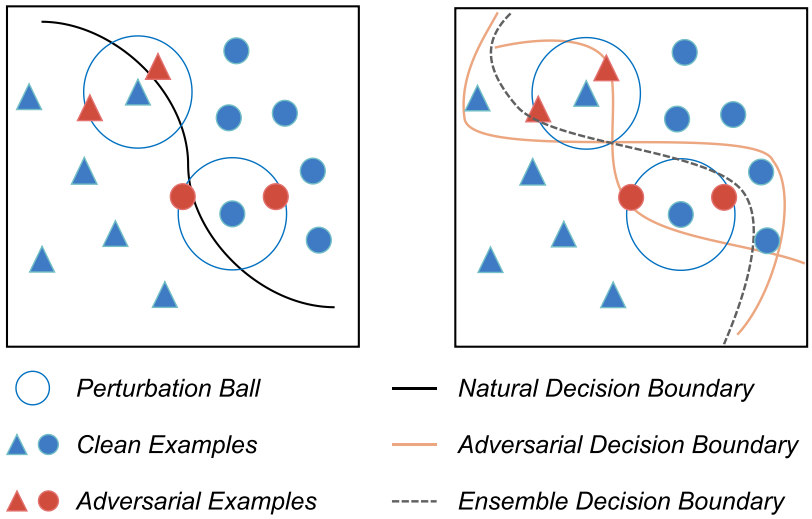
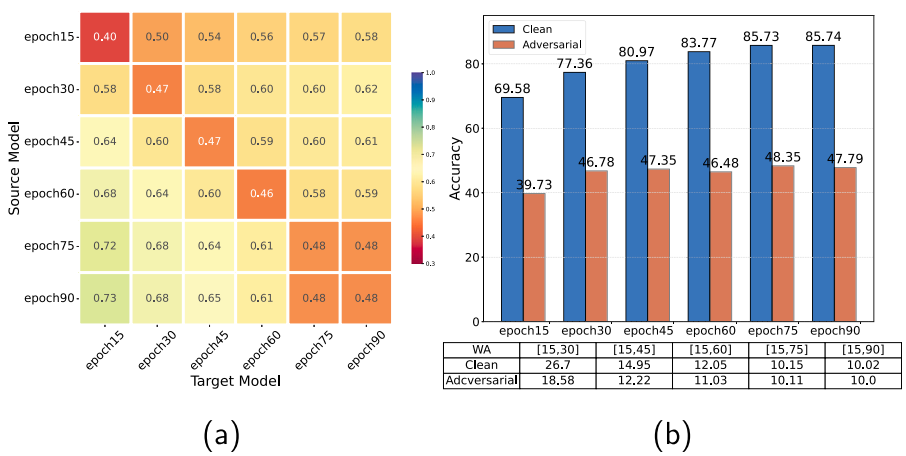


Fig. 1. Schematic of ensemble adversarial decision boundaries and natural decision boundaries. The figure on the left illustrates the decision boundary acquired through training with clean examples. In contrast, the orange lines in the right figure represent different adversarial training periods' decision boundaries. The dashed line indicates the decision boundaries after the ensemble average. It is noteworthy that relying solely on decision boundaries obtained from adversarial examples for ensemble may result in a decline in both robustness and clean accuracy.



WA	[15,30]	[15,45]	[15,60]	[15,75]	[15,90]
Clean	26.7	14.95	12.05	10.15	10.02
Adcversarial	18.58	12.22	11.03	10.11	10.0

Fig. 2. Visualization of clean and adversarial accuracy of member models. (a) shows the adversarial accuracy of the target model for adversarial examples (PGD^{20}) generated on the source model across different epochs. (b) presents the clean and adversarial accuracies for models over each period, including the performance of the ensemble model via weight averaging (WA), with $[15 \times 1,\ 15 \times k]$ representing the ensemble for epochs 15×1 to $15 \times k$. All models are based on standard adversarial training on CIFAR-10, using WRN-34-10. The weakly robust model in the early period of training also has a high adversarial accuracy for the adversarial examples generated on the model in the later training period. For example, when the source model is the model at epoch 60, compared to the target model being the model at epoch 60, the models trained earlier have a higher adversarial accuracy, and the gap between them is more than 10%, which is a huge gap.

called Multi-Guided Adversarial Training (MGAT). MGAT memorizes the weights of the member models at different iterations. Appropriately overfitting the adversarial example is beneficial in increasing the diversity of decision boundaries, it improves the effectiveness of the ensemble [10]. However, excessive overfitting can lead to a decrease in clean accuracy (see Figs. 1 and 2(b)). Therefore, we need to design different guidance strategies when memorizes the weights of the member models.

The guidance strategies mainly address the preservation of diversity in decision boundaries among member models and their alignment with natural decision boundaries. Two types of guidance (*self-guidance* and *expert-guidance*) are used to mitigate the negative impact of interactions between the member models. *Expert-guidance* introduces an expert model to increase this similarity by increasing the logit similarity with the expert model on clean examples. However, simply using this loss may compromise the diversity of decision boundaries. Therefore, *self-guidance* is designed to improve the diversity of decision boundaries. It expands the difference between logit values of correct labels and logit values of incorrect labels from the model itself. The experiments prove that our method is effective in defending against most adversarial attacks. It also illustrates the need to maintain the diversity of decision boundaries of member models while improving their alignment with natural decision boundaries. Our contributions are summarized below:

- We empirically demonstrate that weakly robust models in the early period are still robust to adversarial examples generated on a fully trained robust model. Memorization of the member model weights at different iteration times is necessary.
- We propose a novel ensemble training method called Multi-Guided Adversarial Training (MGAT). The method mitigates the negative impact of interactions between member models on robustness improvement through two guidance approaches (*self-guidance* and *expert-guidance*). These guidance approaches consider maintaining the diversity of the decision boundaries of the member models and simultaneously improving their alignment with natural decision boundaries.

- Experimental results in black-box and white-box settings demonstrate the superiority of our methods in terms of clean accuracy and adversarial accuracy.

2. Related work

We briefly introduce some commonly used attack methods, traditional adversarial training methods, and ensemble adversarial training methods.

2.1. Adversarial attack

Recent studies have shown that high-performance classifiers can still be deceived by adversarial examples [7,15–17]. Thus, adversarial attacks have attracted a great deal of attention since their emergence, giving rise to a large number of methods for generating adversarial examples, which can help assess the robustness of different models and improve their robustness through adversarial training. Here we present some of the most commonly used attack methods as follows.

Goodfellow et al. [2] attributed the existence of adversarial examples to the linear nature of the network, which led to the Fast Gradient Sign Method (FGSM) for efficiently generating adversarial examples. FGSM further evolved into different versions of iterative attack methods. The study in [18] demonstrated the existence of adversarial examples with I-FGSM attacks in the physical world. And that FGSM can be applied in small steps through multiple iterations. Madry et al. [7] found that the I-FGSM can be significantly improved by starting at a random point in the range ball, proposing an attack method called PGD. The CW attack was proposed by [10], which takes into account all three distance metrics to construct strong attacks. AutoAttack (AA) [19] combines a variety of optimized adversarial attack algorithms to construct strong attacks and is widely used to evaluate model robustness. There are also approaches [20–23] that explore the transferability of adversarial examples. Without proper defense measures, it may lead to serious security issues.

2.2. Adversarial training

Adversarial training stands out as one of the most effective defenses against adversarial examples. Methods for improving the robustness of deep learning models through adversarial training can be categorized into three main groups: (1) input-oriented methods, (2) loss-oriented methods, and (3) ensemble-based methods.

A large amount of work has demonstrated the effectiveness of the first type of approach. The study in [24] utilizes a GAN framework to create adversarial examples through a generative model. Zhang et al. [25] controls the length of searching trajectories of PGD through the lens of geometry, In [26], the images synthesized by generative models is used to enhance adversarial training to improve the robustness of adversarial attacks. Wang et al. [27] utilize better generative models (e.g., the popular diffusion model [28]) that can achieve stronger robustness.

For the loss-oriented methods, a series of works have been carried out to establish a new supervised loss for better optimization. Zhang et al. [29] enhances the robustness of the model by minimizing the difference between the original and adversarial examples. The study in [6] considered logit results for models generated from clean and adversarial examples. They created a regularization term by reducing the distance between logit pairs. Zhong et al. [30] used a margin-based approach that considered the margins between clean and adversarial examples. In [31], Wang et al. enhances the robustness of the model by introducing a momentum term, and considering cases where the training data was initially classified correctly or incorrectly, and designing two regularizers for each case. The study in [32] improves Model Robustness by constraining the robust model's logits and making it similar to the logits of the clean model. Xu et al. [33] improves model robustness by encouraging the decision boundary to prioritize points that are susceptible as it moves.

Unlike the traditional improvement, some ensemble methods have also been incorporated into adversarial training. In [34], a regular term is introduced to enrich the variability of the base-model decision boundary to improve the ensemble model's robustness, and SEAT [11] jointly optimizes the state of each historical model on the trajectory during adversarial training.

3. Methodology

In this section, we initially present our motivation (in Section 3.1), followed by the definition of notations and a detailed exploration of specific methods for weight ensemble (in Section 3.2). Ultimately, we introduce our innovative approach, Multi-Guided Adversarial Training (MGAT) (in Section 3.3).

3.1. Motivation

To enhance the robustness of adversarial training through ensemble training, we must address two challenges. (1) The resource issue arises as the individual adversarial training of member models imposes constraints on computational and storage resources. (2) Alleviating the adverse effects resulting from interactions among the member models to effectively improve clean and adversarial accuracy.

Considering the first challenge, we use the weight ensemble instead of ensembling the model result to solve the resource constraints. Combining the phenomena observed in Fig. 2(a), we find that the member model in the early period is still robust to the adversarial examples generated on the member model in the later period. Therefore, when ensemble weights, we should ensure retention of the weights from the latest trained models, while also preserving those from models at earlier stages.

Considering the second challenge, we need to mitigate the impact of interactions between member models on robustness improvement. This interaction is categorized into two situations: (1) If the decision boundaries of the member models are too similar, the ensemble model may not perform as well as the member models. (2) If the decision boundaries of the member models are too different, it may result in the influence of the member models on the ensemble model canceling each other out, thus causing a decrease in the clean accuracy (see Fig. 1). Fig. 2(b) shows that the adversarial and clean accuracies of the member models of the neighboring epochs are somewhat similar. It implies that these models may have similar decision boundaries. Since DNN is no longer a weak classifier in most cases, this is not conducive to improving the effectiveness of the ensemble model. Therefore, we hope to enhance the ensemble effect by improving the diversity of decision boundaries among member models, all the while ensuring their alignment with natural decision boundaries.

To address these two challenges, we propose a novel weight ensemble adversarial training, namely MGAT. This method mitigates the negative impact of interactions between member models on accuracy enhancement through two types of guidance (*self-guidance* and *expert-guidance*). And it enhances the historical memory of member models' weight.

3.2. Notations and weight ensemble

In this section, we will define the notations and give specific methods for weight ensemble. Let $D = \{(x, y)\}$ denote a dataset, where x represents a clean example, and y corresponds to its ground-truth label. The DNN classifier is symbolically represented as $F_\theta(\cdot)$, where θ indicates all trainable parameters. Considering a scenario with L classes, the output vector $F_\theta(x) \in \mathbb{R}^L$. An adversarial example x' is a clean example x added by a tiny well-designed perturbation δ, *i.e.*, $x' = x + \delta$, where the classifier would give a wrong prediction with high

confidence, but human will not be fooled. Subsequently, we provide definitions for two crucial concepts, the *Candidate Weight Pool* and the *Member Model*, essential components employed in the process of weight ensemble.

Definition 1 (*Candidate Weight Pool*)**.** Let $\Theta = \{\theta_0, \theta_1, \dots, \theta_{T-1}\}$ denote the candidate weight pool, where θ_i $(i = 0, \dots, T-1)$ represents the model parameters at the ith iteration, and T denotes the total number of iteration steps.

In the context of weight ensemble, the selection of weights is exclusively from the *Candidate Weight Pool*. To establish a more explicit connection between the weights and the member models, we provide a specific definition of the member model.

Definition 2 (*Member Model*)**.** The member model is the network parameterized by θ_i obtained by the optimization algorithm at each training step when training the network. The member models can be expressed as a sequence $\mathcal{F} = \{F_{\theta_0}, F_{\theta_1}, \dots, F_{\theta_{T-1}}\}$, where F_{θ_i} denotes the deep neural network classifier with a fixed structure parameterized by θ_i.

Recall the goal of adversarial training is to construct a robust model that can effectively generalize to the examples with slight perturbations δ. This objective can be formulated as a min–max problem [35]:

$$\min_{\theta} \max_{\|\delta\|_p < \epsilon} \mathcal{L}(x+\delta, \theta), \tag{1}$$

where ϵ represents the perturbation budget and $\|\cdot\|_p$ represents the p-norm. Internal maximization tries to find an adversarial perturbation that maximizes the loss, while external minimization aims to obtain a robust classifier to adversarial examples [7].

The PGD attack [7] is usually used to solve the internal maximization problem. For each batch, adversarial examples are re-generated for training using PGD attack on these clean examples [7]. This contributes to augmenting the diversity of decision boundaries among member models. When the weight ensemble's smallest unit is an iteration, the decision boundaries acquired by the member models exhibit greater diversity. This stems from the varied clean examples employed in each batch throughout the iterations, introducing randomness to the distribution of generated adversarial examples. However, when considering an epoch as the smallest unit, there is a risk that member models may learn comparable feature representations with similar decision boundaries. This is attributed to the similarity in feature representations among adversarial examples generated from the same clean dataset [12,13,36]. Such similarity is not conducive to enhancing the overall effectiveness of the ensemble model.

Hence, we set the smallest unit for the ensemble as an iteration. The diversity in member model weights is achieved through the randomness introduced by the selection of clean examples for each batch and the variability in the generated adversarial examples.

After addressing the diversity of decision boundaries, we need to consider how to ensemble the weights. A natural consideration is to ensemble by simple averaging: $\tilde{\theta} = \sum_{t=0}^{T-1} \alpha_t \theta_t, s.t. \sum_{t=0}^{T-1} \alpha_t = 1$. However, such a simple average suffers from two problems, one is that it does not capture the latest changes sensitively, and the other is that it may lead to a decrease in clean accuracy. We use the exponential moving average (EMA) from [11] to compute the optimal weight:

$$\tilde{\theta}_{\mathrm{T}} = \alpha \tilde{\theta}_{\mathrm{T}-1} + (1-\alpha)\tilde{\theta}_{\mathrm{T}}, \tag{2}$$

where α is the memory factor. As α gets larger, the degree of memorization of historical weights increases. EMA not only utilizes the latest recommendations of the current optimization but also retains the effects of the previous weighting states. However, in practical experiments, fixing α can lead to poor performance of the ensemble model. The constant c in the Algorithm 1 adjusts the memory factor during the early stages of training. Initially, the model's clean and adversarial accuracies are low. By introducing c, we ensure that a smaller memory factor is used initially, preventing excessive weight integration from poorly trained models and thus affecting the performance of the ensemble model.

3.3. Multi-guided adversarial training

Recall that our objective is to bolster the clean and adversarial accuracy of the ensemble model by amplifying the decision boundary diversity of the member models while concurrently ensuring their alignment with natural decision boundaries. In Section 3.2, we elaborate on the specific methods detailing how to ensure decision boundary diversity and weight ensemble for member models. In this section, we expound on how we enhance the similarity between the decision boundaries of the member models and the natural decision boundaries. Additionally, we introduce the specific algorithmic process of our proposed MGAT.

A common strategy to augment this similarity involves designing a loss associated with clean examples, as employed by [29,31]. However, experimental findings in [11] demonstrated that these loss functions exhibit limited effectiveness in improving the robustness of ensemble models. One potential explanation is that these losses reinforce the similarity between their decision boundaries and natural decision boundaries while compromising the diversity of decision boundaries among member models. Therefore, we employ two specific loss constraints to achieve our goal: the expert-guidance Loss $\mathcal{L}_{SG}$ and the self-guidance Loss $\mathcal{L}_{EG}$.

Given that each iteration of standard adversarial training [7] adjusts decision boundaries based on adversarial examples and subsequently regenerates adversarial examples in the next iteration, the decision boundaries naturally exhibit diversity due to the different malicious inputs. However, this process can lead to significant changes in the logits of the member model for clean examples.

To counterbalance this, we introduce an expert model $H(\cdot)$ obtained through clean training. We aim to enhance the similarity between decision boundaries and clean examples by minimizing the outputs of the member model and the Expert model. Theoretically, expert-guidance Loss in our method refers to any loss function that helps the model better learn the genuine distribution of natural examples during adversarial training. We use Mean-Square Error to measure this similarity, as it mitigates issues with gradient instability that can arise from the sensitivity of loss functions like KL divergence to small probability differences. We define this measure as the expert-guidance Loss $\mathcal{L}_{EG}$:

$$\mathcal{L}_{EG} = \mathbb{E}_{\mathbb{P}}\left[\left(H(x) - F_\theta(x)\right)^2\right], \tag{3}$$

where $\mathbb{P}$ is the probability distribution of the clean examples, $F_\theta(x) \in \mathbb{R}^L$, $H(x) \in \mathbb{R}^L$.

However, simply using this loss may compromise the diversity of decision boundaries. Adversarial examples are a natural consequence of the model's learning of decision boundary generation [7,37]. This property and the complexity of deep neural networks result in decision boundaries generated by adversarial training being unsmooth and diverse [38,39]. Additionally, the diversity of adversarial examples contributes to the diversity of robust decision boundaries. In contrast, the natural decision boundaries of fixed expert models are not diverse. In this context, only using the loss $\mathcal{L}_{EG}$ as a regularization term causes the decision boundaries to align more closely with those obtained from training on natural examples, potentially limiting the diversity of decision boundaries and thus impairing the model's performance on adversarial examples.

Therefore, we start with the member model itself and improve the diversity of decision boundaries. Recall that the original purpose of designing Self-Guidance loss is to give the model the ability to adjust the decision boundaries by itself based on the adversarial examples, while, at the same time, not affecting the diversity of the decision boundaries. To improve the adversarial accuracy, we can expand the difference between the logit values of correct labels and the logit values of incorrect labels. Thus, we adjust the decision boundary by

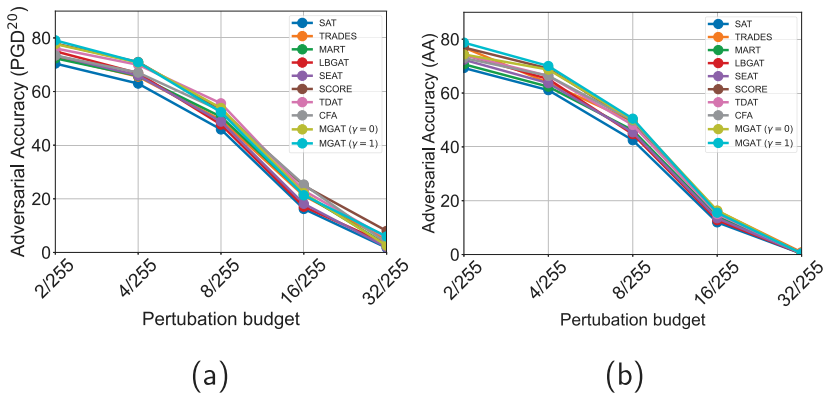


Fig. 3. The adversarial accuracy of each method on ResNet18 with different perturbation budgets in white-box setting. As can be seen from the figure, our method outperforms other methods in dealing with adversarial samples with small perturbations.

Algorithm 1 Multi-Guided Adversarial Training

Input: A DNN classifier $F_{\theta_0}(\cdot)$ with initial learnable parameters θ_0, and A naturally trained expert model $H(\cdot)$, datasets $\mathcal{D}$.
Parameter: number of iterations T; learning rate τ, memory factor α, constant c, smoothing coefficient η, trade-off parameter β, γ.

Let $\theta \leftarrow \theta_0$, $\tilde{\theta} \leftarrow \theta$.
for $t = 0, 1, \cdots, T-1$ **do**
  Sample a minibatch (x, y) from dataset $\mathcal{D}$;
  Generate adversarial examples x' using PGD attack based on x ;
  Compute the loss components:
  $\mathcal{L}_{SG} \leftarrow \mathbb{E}_{\mathbb{P}}\left[\left[F_\theta\left(x'\right)\right]_y - \left(\log \sum_{\hat{y}\neq y} \eta \cdot \exp\left(F_\theta\left(x'\right)\right)\right)/\eta\right]$
  $\mathcal{L}_{EG} \leftarrow \mathbb{E}_{\mathbb{P}}\left[\left(H(x) - F_\theta(x)\right)^2\right]$
  Compute total loss: $\mathcal{L} \leftarrow \mathcal{L}_{CE} + \beta\mathcal{L}_{SG} + \gamma\mathcal{L}_{EG}$;
  $g_{\theta_t} \leftarrow \mathbb{E}_{(x',y)}[\nabla_{\theta_t}\mathcal{L}(\theta_t; (x', y))]$ in every minibatch;
  Calculate τ_t according to the current iterations;
  $\theta_t \leftarrow \theta_t - \tau g_{\theta_t}$;
  $\alpha' \leftarrow \min\left(\alpha, \frac{t}{t+c}\right)$;
  $\tilde{\theta} \leftarrow \alpha_t\tilde{\theta} + (1-\alpha_t)\theta_t$;
end for
return A emsemble adversarial training model $F_{\tilde{\theta}}(\cdot)$.

controlling the margin of adversarial examples, defining self-guidance loss $\mathcal{L}_{SG}$:

$$\mathcal{L}_{SG} = -\mathbb{E}_{\mathbb{P}}\left[\left[F_\theta\left(x'\right)\right]_y - \left(\log \sum_{\hat{y}\neq y} \eta \cdot \exp\left(F_\theta\left(x'\right)\right)\right)/\eta\right], \tag{4}$$

where $\hat{y}$ is the predicted label of $F_\theta(\cdot)$ for x'. The parameter η serves as the smoothing coefficient, where η is constrained to the interval $(0, 1]$. A smaller value of η imparts more smoothness to the model concerning the logit distribution of x. Conversely, a larger η accentuates the distinction between the logit values corresponding to correct labels and those associated with incorrect labels. Minimizing this loss enables the model to adjust the decision boundary to correctly classify adversarial examples. This adjustment continues even when the predicted label is correct, thus increasing the diversity of decision boundaries. The detailed flow of the algorithm is shown in algorithm 1.

4. Experiments

4.1. Setups

We conducted extensive experiments in both white-box and black-box settings, utilizing the CIFAR-10 [40], CIFAR-100 [40] and Tiny-ImageNet [41] without incorporating additional data.

4.1.1. Defense settings

We employed WideResNet-34-10 (WRN-34-10) [42] and ResNet-18 [43] as default models, utilizing SGD optimizer with a momentum of 0.9 and weight decay of 2×10^{-4}. The models were trained for 100 epochs with a batch size of 128. For ResNet-18, the initial learning rate was set to 0.01, while for WRN-34-10, it was set to 0.1. The learning rate decayed to one-tenth at the 75th and 90th epochs. PGD^{10} was employed to generate adversarial examples for training. We compare MGAT with the following strong baseline models: SAT [7], TRADES [29], MART [31], LBGAT [32], SEAT [11], SCORE [44], TDAT [45], and CFA [46]. Except for the mentioned settings, all parameters of the baseline methods adhere to the original configurations. For MGAT, the expert model is the clean-trained model that shares the same structure as the training model. We set $\beta = 1$, $\gamma = 1$, $\eta = 0.995$, $c = 10$, and $\alpha = 0.999$.

4.1.2. Attack settings

We primarily employ adversarial attacks within the $l\infty$ constraints. The evaluation of the adversarial model's robustness in the white-box setting relied on methods such as PGD [7], MI-FGSM [20], CW [10], and AA [19]. In the black-box scenario, we choose the adversarially trained VGG-16 [47] as the surrogate model, generate adversarial examples on the surrogate model, and then calculate the accuracy of each method. FGSM [2], PGD^{20} [7], TI-FGSM [21], MI-FGSM [20], DI-FGSM [22], and VMI-FGSM [23] are used to assess model performance. The perturbation budget for all the above attacks was set at 0.031. PGD^K denoted a PGD attack with K iteration steps and a step size of 0.007.

4.2. Performance under white-box attacks

In white-box settings, all attacks have access to the architecture and parameters of the target model. There are two training methods for MGAT: when $\gamma = 0$, MGAT primarily involves *self-guidance*, and when $\gamma = 1$, MGAT incorporates both *expert-guidance* and *self-guidance*, enhancing the model's accuracy further. Results in Tables 1 and 2 reveal that the clean and adversarial accuracies of MGAT ($\gamma = 1$) surpass significantly those of other methods. Notably, the clean accuracy of MGAT ($\gamma = 1$) outperforms other methods significantly, with a 1.37% improvement in clean accuracy compared to the second-highest method in the model results of ResNet-18 trained on the CIFAR-10. The improvement is even more pronounced in ResNet-18 compared with WRN-34-10, which may be attributed to ResNet-18's simpler model structure, allowing for more diverse decision boundaries.

However, the model trained with MGAT ($\gamma = 1$) is less effective under adversarial attacks such as PGD^{20}, PGD^{100}, and MI-FGSM compared to MGAT ($\gamma = 0$) in CIFAR-10 and CIFAR-100. One potential explanation is that $\mathcal{L}_{EG}$ diminishes the diversity of decision boundaries

Table 1
Adversarial accuracy (%) of adversarial training against white-box attacks. The target model is ResNet-18. The best results are indicated in bold.

Dataset	Defenses	None	PGD^{20}	PGD^{100}	MI-FGSM	CW	AA
CIFAR-10	SAT [7]	77.59 ± 0.62	45.45 ± 0.35	45.42 ± 0.42	46.32 ± 0.34	43.90 ± 0.63	42.93 ± 0.41
	TRADES [29]	80.52 ± 0.88	49.41 ± 0.44	49.40 ± 0.38	50.02 ± 0.09	47.17 ± 0.46	46.06 ± 0.94
	MART [31]	79.97 ± 0.62	50.59 ± 0.01	50.58 ± 0.02	50.43 ± 0.01	47.39 ± 0.60	46.18 ± 0.18
	LBGAT [32]	82.42 ± 0.69	47.69 ± 0.07	47.72 ± 0.13	48.60 ± 0.19	45.68 ± 0.06	44.73 ± 0.11
	SEAT [11]	79.91 ± 0.28	48.79 ± 0.02	48.80 ± 0.02	49.19 ± 0.57	47.42 ± 0.02	45.39 ± 0.01
	SCORE [44]	84.11 ± 0.18	53.14 ± 0.28	52.91 ± 0.30	53.37 ± 0.17	50.59 ± 0.21	49.36 ± 0.37
	TDAT [45]	81.26 ± 0.63	**55.59 ± 0.93**	**55.64 ± 0.84**	**56.22 ± 0.78**	49.76 ± 0.31	48.13 ± 0.27
	CFA [46]	76.51 ± 0.81	52.83 ± 0.47	52.91 ± 0.44	53.00 ± 0.51	48.85 ± 0.19	49.76 ± 0.28
	MGAT ($\gamma = 0$) **(ours)**	83.34 ± 0.25	53.85 ± 0.04	53.81 ± 0.01	54.68 ± 0.25	52.30 ± 0.34	50.13 ± 0.18
	MGAT ($\gamma = 1$) **(ours)**	**85.48 ± 0.53**	52.31 ± 0.18	52.29 ± 0.19	53.61 ± 0.22	**53.07 ± 0.14**	**50.48 ± 0.23**
CIFAR-100	SAT [7]	51.83 ± 0.47	22.55 ± 0.13	22.56 ± 0.16	23.21 ± 0.02	21.26 ± 0.01	19.32 ± 0.03
	TRADES [29]	53.46 ± 0.86	25.98 ± 1.42	26.06 ± 1.38	26.32 ± 1.41	21.90 ± 1.62	21.96 ± 0.49
	MART [31]	51.35 ± 0.02	26.28 ± 0.03	26.26 ± 0.02	26.69 ± 0.01	23.13 ± 0.01	21.81 ± 0.02
	LBGAT [32]	56.85 ± 1.12	29.91 ± 0.81	29.88 ± 0.88	30.12 ± 0.78	24.77 ± 0.57	25.50 ± 0.46
	SEAT [11]	53.74 ± 0.88	26.25 ± 0.08	26.21 ± 0.04	26.77 ± 0.13	24.22 ± 0.27	22.49 ± 0.21
	SCORE [44]	55.99 ± 0.57	29.71 ± 0.09	29.56 ± 0.14	29.89 ± 0.10	26.16 ± 0.31	24.67 ± 0.33
	TDAT [45]	**64.12 ± 0.51**	17.99 ± 1.43	18.05 ± 1.39	14.01 ± 1.28	14.18 ± 0.60	14.10 ± 0.51
	CFA [46]	57.39 ± 1.24	29.12 ± 0.67	28.02 ± 0.66	28.08 ± 0.24	26.71 ± 0.31	25.13 ± 0.42
	MGAT ($\gamma = 0$) **(ours)**	58.83 ± 1.18	**30.83 ± 0.23**	**30.84 ± 0.23**	**31.46 ± 0.13**	**28.73 ± 0.04**	**26.60 ± 0.08**
	MGAT ($\gamma = 1$) **(ours)**	62.46 ± 1.09	28.16 ± 0.09	28.21 ± 0.01	29.04 ± 0.04	28.16 ± 0.11	25.54 ± 0.10
Tiny-ImageNet	SAT [7]	30.34 ± 0.73	12.20 ± 0.45	12.26 ± 0.41	12.36 ± 0.33	10.00 ± 0.07	8.44 ± 0.49
	TRADES [29]	36.19 ± 0.66	11.89 ± 0.08	11.86 ± 0.08	11.96 ± 0.04	8.51 ± 0.06	8.26 ± 0.21
	MART [31]	33.57 ± 0.36	13.93 ± 1.13	13.93 ± 1.14	14.07 ± 1.14	11.25 ± 0.92	10.39 ± 0.77
	LBGAT [32]	36.68 ± 0.79	10.62 ± 0.27	10.70 ± 0.26	11.08 ± 0.17	10.12 ± 0.12	8.12 ± 0.11
	SEAT [11]	32.34 ± 0.62	13.70 ± 0.28	13.72 ± 0.32	13.80 ± 0.09	11.04 ± 0.25	9.64 ± 0.34
	SCORE [44]	12.53 ± 0.41	7.32 ± 0.49	7.32 ± 0.57	7.40 ± 0.59	5.72 ± 0.26	5.30 ± 0.14
	TDAT [45]	21.04 ± 0.68	13.86 ± 0.96	14.00 ± 0.99	14.02 ± 0.83	11.24 ± 0.37	9.74 ± 0.38
	CFA [46]	36.48 ± 1.17	13.00 ± 0.87	13.02 ± 0.78	13.32 ± 0.81	11.36 ± 0.69	10.04 ± 0.50
	MGAT ($\gamma = 0$) **(ours)**	33.42 ± 1.33	12.65 ± 0.97	12.62 ± 0.97	12.92 ± 0.64	10.41 ± 0.84	9.80 ± 0.50
	MGAT ($\gamma = 1$) **(ours)**	**38.01 ± 0.31**	**14.90 ± 0.64**	**14.92 ± 0.69**	**14.72 ± 0.34**	**12.94 ± 0.71**	**10.48 ± 0.22**

Table 2
Adversarial accuracy (%) of adversarial training against white-box attacks. The target model is WRN-34-10. The best results are indicated in bold.

Dataset	Defenses	None	PGD^{20}	PGD^{100}	MI-FGSM	CW	AA
CIFAR-10	SAT [7]	85.61 ± 0.04	46.67 ± 0.03	46.74 ± 0.04	48.05 ± 0.20	47.01 ± 0.15	44.56 ± 2.04
	TRADES [29]	85.45 ± 0.12	51.89 ± 2.00	51.01 ± 2.08	53.41 ± 1.07	52.10 ± 0.14	50.16 ± 1.77
	MART [31]	83.70 ± 0.24	53.96 ± 0.09	53.98 ± 0.11	55.14 ± 0.14	51.31 ± 0.22	49.44 ± 0.03
	LBGAT [32]	80.50 ± 0.18	55.84 ± 0.42	55.85 ± 0.33	56.45 ± 0.13	53.81 ± 0.15	52.85 ± 0.32
	SEAT [11]	84.35 ± 0.30	52.73 ± 0.12	52.71 ± 0.13	53.82 ± 0.10	51.96 ± 0.08	50.05 ± 0.05
	SCORE [44]	85.22 ± 0.39	55.62 ± 0.16	55.49 ± 0.15	56.15 ± 0.38	53.77 ± 0.46	52.70 ± 0.11
	TDAT [45]	**92.85 ± 1.97**	47.50 ± 1.23	47.04 ± 1.21	49.74 ± 0.88	39.80 ± 0.74	24.53 ± 0.64
	CFA [46]	83.62 ± 0.91	56.34 ± 0.98	56.32 ± 0.97	57.41 ± 0.45	54.97 ± 0.31	52.30 ± 0.13
	MGAT ($\gamma = 0$) **(ours)**	86.70 ± 0.06	**57.87 ± 0.12**	**57.93 ± 0.11**	**58.86 ± 0.04**	56.72 ± 0.16	54.63 ± 0.02
	MGAT ($\gamma = 1$) **(ours)**	88.01 ± 0.04	57.18 ± 0.11	57.21 ± 0.21	58.41 ± 0.27	**58.05 ± 0.14**	**55.48 ± 0.21**
CIFAR-100	SAT [7]	53.64 ± 0.47	24.49 ± 0.13	24.48 ± 0.16	24.97 ± 0.02	23.00 ± 0.01	21.01 ± 0.01
	TRADES [29]	58.54 ± 0.14	29.92 ± 1.25	29.93 ± 1.04	30.27 ± 0.99	27.21 ± 0.39	25.96 ± 0.01
	MART [31]	57.99 ± 0.25	28.89 ± 0.18	28.87 ± 0.20	29.43 ± 0.13	27.10 ± 0.28	25.74 ± 0.25
	LBGAT [32]	59.98 ± 0.43	33.42 ± 0.27	33.41 ± 0.26	33.80 ± 0.18	33.20 ± 0.31	29.06 ± 0.14
	SEAT [11]	57.53 ± 0.39	29.95 ± 0.04	29.96 ± 0.05	30.59 ± 0.04	28.52 ± 0.01	26.17 ± 0.01
	SCORE [44]	56.07 ± 0.34	29.40 ± 0.17	29.37 ± 0.18	29.87 ± 0.22	26.61 ± 0.34	24.98 ± 0.24
	TDAT [45]	43.58 ± 1.50	26.10 ± 1.46	26.02 ± 1.39	26.22 ± 1.09	22.00 ± 0.57	21.01 ± 0.53
	CFA [46]	61.28 ± 1.01	30.67 ± 0.85	30.67 ± 0.77	31.09 ± 0.53	29.21 ± 0.63	30.23 ± 0.10
	MGAT ($\gamma = 0$) **(ours)**	63.38 ± 0.47	**34.20 ± 0.07**	**34.21 ± 0.06**	**34.82 ± 0.06**	32.69 ± 0.02	30.38 ± 0.01
	MGAT ($\gamma = 1$) **(ours)**	**66.00 ± 0.69**	33.31 ± 0.33	33.31 ± 0.24	34.17 ± 0.11	**33.21 ± 0.06**	**30.47 ± 0.10**

among member models. This results in a degradation of the ensemble model's resilience to weak robust attacks (PGD, MI-FGSM). Nevertheless, MGAT ($\gamma = 1$) still maintains a relatively high accuracy under more complex attacks like CW and AA, which shows resistance to sophisticated and refined adversarial attacks. This resistance is also effective under different perturbation budgets (see Fig. 3). These findings underscore the delicate balance required between the diversity of decision boundaries within the member model and their similarity to clean decision boundaries.

4.3. Performance under black-box attacks

In black-box settings, attacks know only part of the attacked model's output when generating adversarial examples. We select the model that performs best in the white-box setting for testing. Tables 3 and 4 showcase the outstanding performance of MGAT across diverse datasets and model structures. While the robustness of MGAT ($\gamma = 0$) has seen an improvement compared to other methods, it still falls significantly short of MGAT ($\gamma = 1$), underscoring the indispensability of expert guidance. The introduction of expert-guidance contributes to enhancing the model's robustness in the black-box setting.

4.4. Ablation

4.4.1. Impact of weight ensemble

We conducted training on WRN-34-10 and generated plots illustrating the evolution of both clean and adversarial accuracy on the test set for each epoch, as shown in Fig. 4. Simultaneously, the data in Fig. 5 underscores the substantial impact of the weight ensemble strategy in enhancing robust accuracy. In contrast to the modest 0.23%

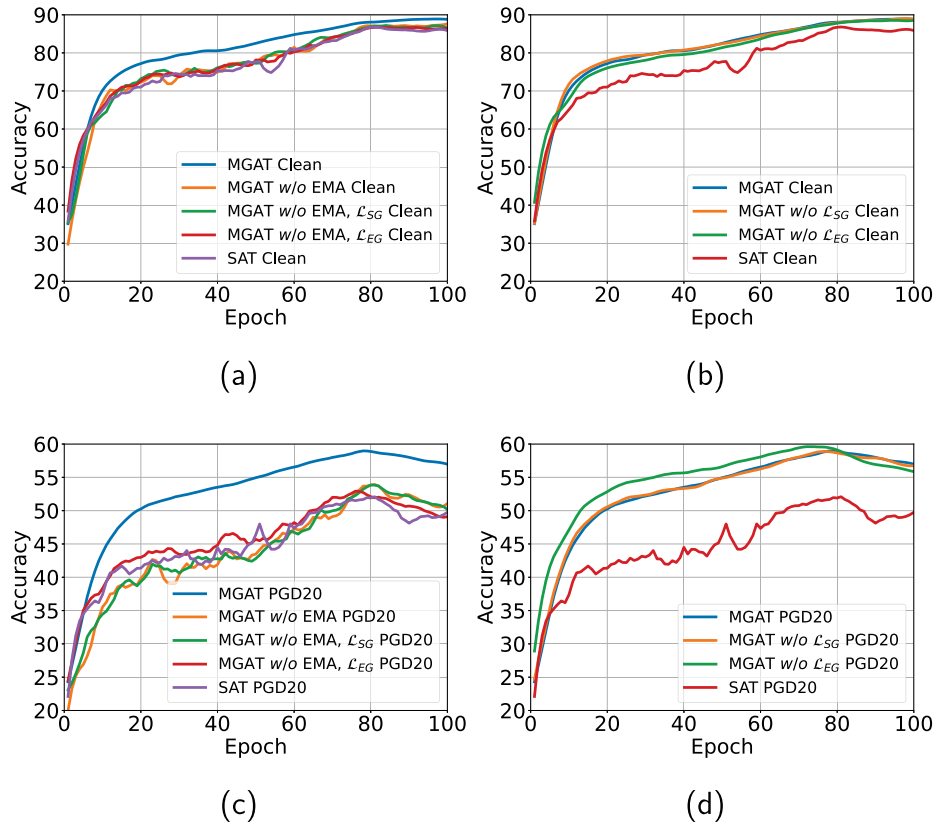


Fig. 4. Variation curves of clean and adversarial accuracy of MGAT. This curve was obtained by training WRN-34-10 on CIFAR-10. It can be seen that all parts of MGAT have positive effects on the improvement of clean and adversarial accuracy.

Table 3
Adversarial accuracy (%) of defense methods against black-box attacks. The target model is ResNet-18. The best results are indicated in bold.

Dataset	Defense	FGSM	PGD^{20}	TI-FGSM	MI-FGSM	DI-FGSM	VMI-FGSM
CIFAR-10	SAT [7]	61.72	61.16	65.92	60.89	61.27	60.48
	TRADES [29]	64.96	64.48	68.68	64.29	64.55	64.07
	MART [31]	63.50	63.00	67.93	62.78	63.19	62.61
	LBGAT [32]	66.72	66.24	71.70	66.05	66.18	65.74
	SEAT [11]	63.75	63.17	68.90	63.09	63.49	62.90
	SCORE [44]	69.31	68.54	74.15	69.57	69.26	69.45
	TDAT [44]	66.51	66.31	71.41	66.03	66.47	65.84
	CFA [46]	61.85	62.11	66.18	61.75	61.77	61.63
	MGAT ($\gamma = 0$) **(ours)**	69.36	69.13	74.50	68.82	69.44	68.64
	MGAT ($\gamma = 1$) **(ours)**	**71.69**	**71.49**	**76.94**	**71.10**	**71.80**	**70.91**
CIFAR-100	SAT [7]	37.40	37.15	40.23	37.02	37.26	36.84
	TRADES [29]	40.07	39.82	42.52	39.08	39.78	39.45
	MART [31]	38.53	38.39	40.51	38.02	38.13	37.92
	LBGAT [32]	42.82	42.95	45.83	42.62	42.69	42.50
	SEAT [11]	38.80	38.39	41.90	38.22	38.35	37.97
	SCORE [44]	42.34	42.47	45.80	42.17	42.46	42.02
	TDAT [45]	48.72	48.47	53.00	48.15	48.26	47.61
	CFA [46]	43.87	43.64	46.67	43.30	43.38	43.22
	MGAT ($\gamma = 0$) **(ours)**	44.21	44.12	48.03	43.99	44.33	43.92
	MGAT ($\gamma = 1$) **(ours)**	**48.88**	**48.75**	**53.13**	**48.62**	**48.77**	**48.44**
Tiny-ImageNet	SAT [7]	28.96	29.26	29.28	29.16	29.20	29.16
	TRADES [29]	35.78	35.81	35.75	35.78	35.79	35.59
	MART [31]	33.33	33.39	33.38	33.35	33.39	33.16
	LBGAT [32]	32.76	34.74	34.40	34.62	34.42	34.36
	SEAT [11]	31.02	31.38	31.34	31.28	31.02	31.38
	SCORE [44]	11.80	11.88	11.76	11.82	11.82	11.82
	TDAT [45]	20.88	20.82	20.74	20.86	20.88	20.86
	CFA [46]	35.16	35.42	35.54	35.26	35.34	35.26
	MGAT ($\gamma = 0$) **(ours)**	31.62	32.16	31.86	32.16	31.76	31.94
	MGAT ($\gamma = 1$) **(ours)**	**36.84**	**37.74**	**37.58**	**37.46**	**37.38**	**37.20**

Table 4
Adversarial accuracy (%) of defense methods against black-box attacks. The target model is WRN-34-10. The best results are indicated in bold.

Dataset	Defense	FGSM	PGD^{20}	TI-FGSM	MI-FGSM	DI-FGSM	VMI-FGSM
CIFAR-10	SAT [7]	73.29	72.75	77.47	72.64	73.07	72.47
	TRADES [29]	74.06	74.01	78.28	74.85	75.11	74.81
	MART [31]	71.45	71.19	75.62	70.90	71.31	70.71
	LBGAT [32]	68.65	68.96	71.80	68.72	68.78	68.59
	SEAT [11]	69.55	69.17	74.57	68.95	69.38	68.77
	SCORE [44]	70.69	70.70	75.37	69.91	70.28	69.74
	TDAT [45]	**82.67**	**82.71**	**89.39**	**82.34**	**83.43**	**82.12**
	CFA [46]	71.94	72.11	76.28	71.76	72.13	71.51
	MGAT ($\gamma = 0$) **(ours)**	74.70	74.86	79.32	74.61	75.02	74.43
	MGAT ($\gamma = 1$) **(ours)**	77.23	77.26	81.47	76.97	77.50	76.91
CIFAR-100	SAT [7]	38.70	38.57	41.69	38.38	38.64	38.24
	TRADES [29]	46.39	46.31	48.48	46.10	46.21	46.07
	MART [31]	46.03	46.13	48.24	45.64	45.84	45.66
	LBGAT [32]	48.45	48.43	50.67	48.16	48.30	48.12
	SEAT [11]	43.82	43.50	46.82	43.42	43.53	43.28
	SCORE [44]	43.76	43.90	46.72	43.50	43.68	43.45
	TDAT [45]	32.86	32.77	34.67	32.59	32.57	32.46
	CFA [46]	49.10	49.14	51.59	48.91	48.90	48.82
	MGAT ($\gamma = 0$) **(ours)**	50.55	50.78	54.15	50.49	50.75	50.38
	MGAT ($\gamma = 1$) **(ours)**	**52.35**	**52.36**	**56.21**	**52.28**	**52.35**	**52.13**

Table 5
Evaluation of MGAT under different parameter settings. All experiments were obtained using the WRN-34-10 model structure trained on CIFAR-10. The performance of the best checkpoint (Best) and the performance of the last checkpoint (Last) are reported.

Settings	Clean Best/Last	PGD^{20} Best/Last	AA Best/Last	Settings	Clean Best/Last	PGD^{20} Best/Last	AA Best/Last
$\beta = 0.0$	87.59/**89.73**	56.36/54.07	54.73/53.29	$\gamma = 0.0$	86.52/87.84	**58.11**/56.91	54.66/54.50
$\beta = 0.3$	87.39/89.24	56.20/54.98	54.68/53.27	$\gamma = 0.3$	87.86/88.53	57.62/**57.04**	55.35/55.27
$\beta = 0.5$	87.65/88.64	56.73/56.01	55.04/54.83	$\gamma = 0.5$	87.89/88.87	57.38/56.94	55.52/55.38
$\beta = 0.7$	87.92/89.01	57.01/56.11	55.36/55.16	$\gamma = 0.7$	87.95/88.74	57.07/57.01	55.36/55.19
$\beta = 1.0$	**88.04**/89.07	**57.26/56.13**	**55.62/55.41**	$\gamma = 1.0$	**88.04/89.07**	57.26/56.13	**55.62/55.41**

improvement in clean accuracy, the adversarial accuracy, particularly on Adversarial Attacks (AA), exhibits a remarkable improvement of almost 5.85%. Compared to its impact on clean accuracy, the weight ensemble proves to be more effective in enhancing adversarial accuracy. In conjunction with Fig. 4, it indicates that one contributing factor is the model's lower accuracy in classifying adversarial examples at different epochs compared to clean examples. What is more, each iteration generates new adversarial examples on the current model, leading to significantly different decision boundaries and, consequently, a pronounced increase in ensemble effects.

4.4.2. Impact of $\mathcal{L}_{EG}$

We elaborate on the details of the model without the inclusion of $\mathcal{L}_{EG}$ (MGAT ($\gamma = 0$)) (refer to Tables 1 and 2). We observed a reduction in clean accuracy when decreasing the tradeoff parameter γ of $\mathcal{L}_{EG}$, MGAT ($\gamma = 0$) exhibits increased robustness against weak attacks in the white-box setting (see Table 5). Simultaneously, there is a decrease in robustness against strong attacks on the CIFAR-10 dataset. These phenomena are manifested at both the best and the last checkpoints. This decline can be attributed to the absence of guidance from the expert model after removing $\mathcal{L}_{EG}$, the model tends to overfit the adversarial examples generated by PGD. We note that this phenomenon is less pronounced on CIFAR-100, likely because CIFAR-100 has more classes and is inherently more challenging to overfit. Therefore, the primary role of $\mathcal{L}_{EG}$ is to enhance clean accuracy and serve as a regularizer when the model is prone to overfit.

4.4.3. Impact of $\mathcal{L}_{SG}$

When $\mathcal{L}_{SG}$ is omitted from the loss, a noticeable degradation in both clean and adversarial accuracies is observed, as presented in Fig. 5. In Fig. 4(c), the curve depicting the model's adversarial accuracy exhibits a steeper change, and Fig. 4(d) illustrates a higher adversarial accuracy after weight ensemble when $\mathcal{L}_{SG}$ is considered. When β decreases, the adversarial accuracy against both PGD and AA decreases (see Table 5). When $\beta = 0$, under the continued action of $\mathcal{L}_{EG}$, the diversity

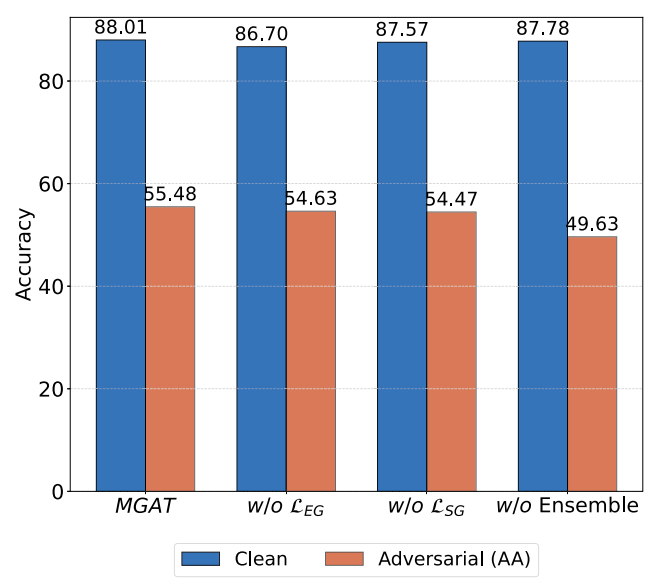


Fig. 5. Results of ablation experiments of MGAT. All ablation experiments were obtained using the WRN-34-10 model structure trained on CIFAR-10.

of decision boundaries is attenuated and the model tends to overfit the adversarial examples generated on the training set in the later stages of training. This is corroborated by the decrease in robustness performance and the rise in clean accuracy at the last checkpoint (see Table 5). These observations collectively highlight the significant role played by $\mathcal{L}_{SG}$ in enhancing the diversity of decision boundaries among the member models.

5. Conclusion

In this paper, we propose a novel weight ensemble method called MGAT. The method mitigates the negative impact of interactions between member models on accuracy improvement through two guidance approaches (*self-guidance* and *expert-guidance*). The experiments prove that our method is effective in defending against most adversarial attacks. In future work, we can extend this guided training approach to contribute to more domains by incorporating the naturally occurring diversity of member models in adversarial training and the interplay of positive and negative member models.

CRediT authorship contribution statement

Chenyu Zhao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Yaguan Qian:** Writing – review & editing, Investigation, Conceptualization. **Bin Wang:** Investigation, Funding acquisition, Conceptualization. **Zhaoquan Gu:** Funding acquisition, Conceptualization. **Shouling Ji:** Investigation, Conceptualization. **Wei Wang:** Funding acquisition, Conceptualization. **Yanchun Zhang:** Investigation, Conceptualization.

Declaration of competing interest

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Data availability

The data used in this study are publicly available.

References

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I.J. Goodfellow, R. Fergus, Intriguing properties of neural networks, in: Int. Conf. Learn. Represent., 2014.
[2] I.J. Goodfellow, J. Shlens, C. Szegedy, Explaining and harnessing adversarial examples, in: Int. Conf. Learn. Represent., 2015.
[3] N. Papernot, P.D. McDaniel, X. Wu, S. Jha, A. Swami, Distillation as a defense to adversarial perturbations against deep neural networks, in: IEEE Symposium on Security and Privacy, SP, IEEE Computer Society, 2016, pp. 582–597.
[4] W. Xu, D. Evans, Y. Qi, Feature squeezing: Detecting adversarial examples in deep neural networks, in: 25th Annual Network and Distributed System Security Symposium, NDSS 2018, San Diego, California, USA, February 18-21, 2018, The Internet Society, 2018.
[5] C. Xie, J. Wang, Z. Zhang, Z. Ren, A.L. Yuille, Mitigating adversarial effects through randomization, in: Int. Conf. Learn. Represent., 2018.
[6] H. Kannan, A. Kurakin, I.J. Goodfellow, Adversarial logit pairing, 2018, CoRR abs/1803.06373.
[7] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, A. Vladu, Towards deep learning models resistant to adversarial attacks, in: Int. Conf. Learn. Represent., 2018.
[8] F. Tramèr, A. Kurakin, N. Papernot, I.J. Goodfellow, D. Boneh, P.D. McDaniel, Ensemble adversarial training: Attacks and defenses, in: Int. Conf. Learn. Represent., 2018.
[9] X. Liang, Y. Qian, J. Huang, X. Ling, B. Wang, C. Wu, W. Swaileh, Towards desirable decision boundary by moderate-margin adversarial training, in: Pattern Recognit. Lett., 173, 2023, pp. 30–37.
[10] N. Carlini, D.A. Wagner, Towards evaluating the robustness of neural networks, in: 2017 IEEE Symposium on Security and Privacy, SP, IEEE Computer Society, 2017, pp. 39–57.
[11] H. Wang, Y. Wang, Self-ensemble adversarial training for improved robustness, in: Int. Conf. Learn. Represent., 2022.
[12] Y.N. Dauphin, R. Pascanu, Ç. Gülçehre, K. Cho, S. Ganguli, Y. Bengio, Identifying and attacking the saddle point problem in high-dimensional non-convex optimization, in: Adv. Neural Inform. Process. Syst., 2014, pp. 2933–2941.
[13] Y. Li, J. Yosinski, J. Clune, H. Lipson, J.E. Hopcroft, Convergent Learning: Do different neural networks learn the same representations? in: Int. Conf. Learn. Represent., 2016.
[14] Y. Balaji, T. Goldstein, J. Hoffman, Instance adaptive adversarial training: Improved accuracy tradeoffs in neural nets, 2019, CoRR abs/1910.08051.
[15] A. Kurakin, I.J. Goodfellow, S. Bengio, Y. Dong, F. Liao, M. Liang, T. Pang, J. Zhu, X. Hu, C. Xie, J. Wang, Z. Zhang, Z. Ren, A.L. Yuille, S. Huang, Y. Zhao, Y. Zhao, Z. Han, J. Long, Y. Berdibekov, T. Akiba, S. Tokui, M. Abe, Adversarial attacks and defences competition, 2018, CoRR abs/1804.00097.
[16] Y. Qian, S. He, C. Zhao, J. Sha, W. Wang, B. Wang, Lea^2: A lightweight ensemble adversarial attack via non-overlapping vulnerable frequency regions, in: Int. Conf. Comput. Vis., IEEE, 2023, pp. 4487–4498.
[17] Y. Qian, K. Chen, B. Wang, Z. Gu, S. Ji, W. Wang, Y. Zhang, Enhancing transferability of adversarial examples through mixed-frequency inputs, in: IEEE Trans. Inf. Forensics Secur., 19, 2024, pp. 7633–7645.
[18] A. Kurakin, I.J. Goodfellow, S. Bengio, Adversarial machine learning at scale, in: Int. Conf. Learn. Represent., 2017.
[19] F. Croce, M. Hein, Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks, in: Int. Conf. Learn. Represent., in: Proceedings of Machine Learning Research, Vol. 119, PMLR, 2020, pp. 2206–2216.
[20] Y. Dong, F. Liao, T. Pang, H. Su, J. Zhu, X. Hu, J. Li, Boosting adversarial attacks with momentum, in: IEEE Conf. Comput. Vis. Pattern Recog., Computer Vision Foundation / IEEE Computer Society, 2018, pp. 9185–9193.
[21] Y. Dong, T. Pang, H. Su, J. Zhu, Evading defenses to transferable adversarial examples by translation-invariant attacks, in: IEEE Conf. Comput. Vis. Pattern Recog., Computer Vision Foundation / IEEE, 2019, pp. 4312–4321.
[22] C. Xie, Z. Zhang, Y. Zhou, S. Bai, J. Wang, Z. Ren, A.L. Yuille, Improving transferability of adversarial examples with input diversity, in: IEEE Conf. Comput. Vis. Pattern Recog., Computer Vision Foundation / IEEE, 2019, pp. 2730–2739.
[23] X. Wang, K. He, Enhancing the transferability of adversarial attacks through variance tuning, in: IEEE Conf. Comput. Vis. Pattern Recog., Computer Vision Foundation / IEEE, 2021, pp. 1924–1933.
[24] S. Baluja, I. Fischer, Adversarial transformation networks: Learning to generate adversarial examples, 2017, CoRR abs/1703.09387.
[25] J. Zhang, J. Zhu, G. Niu, B. Han, M. Sugiyama, M.S. Kankanhalli, Geometry-aware instance-reweighted adversarial training, in: Int. Conf. Learn. Represent., 2021.
[26] S. Gowal, S. Rebuffi, O. Wiles, F. Stimberg, D.A. Calian, T.A. Mann, Improving robustness using generated data, in: Adv. Neural Inform. Process. Syst., 2021, pp. 4218–4233.
[27] Z. Wang, T. Pang, C. Du, M. Lin, W. Liu, S. Yan, Better diffusion models further improve adversarial training, in: Int. Conf. Learn. Represent., in: Proceedings of Machine Learning Research, Vol. 202, PMLR, 2023, pp. 36246–36263.
[28] T. Karras, M. Aittala, T. Aila, S. Laine, Elucidating the design space of diffusion-based generative models, in: Adv. Neural Inform. Process. Syst., 2022.
[29] H. Zhang, Y. Yu, J. Jiao, E.P. Xing, L.E. Ghaoui, M.I. Jordan, Theoretically principled trade-off between robustness and accuracy, in: Int. Conf. Learn. Represent., in: Proceedings of Machine Learning Research, Vol. 97, PMLR, 2019, pp. 7472–7482.
[30] Y. Zhong, W. Deng, Adversarial learning with margin-based triplet embedding regularization, in: Int. Conf. Comput. Vis., IEEE, 2019, pp. 6548–6557.
[31] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, Q. Gu, Improving adversarial robustness requires revisiting misclassified examples, in: Int. Conf. Learn. Represent., 2020.
[32] J. Cui, S. Liu, L. Wang, J. Jia, Learnable boundary guided adversarial training, in: Int. Conf. Comput. Vis., IEEE, 2021, pp. 15701–15710.
[33] Y. Xu, Y. Sun, M. Goldblum, T. Goldstein, F. Huang, Exploring and exploiting decision boundary dynamics for adversarial robustness, in: Int. Conf. Learn. Represent., 2023.
[34] T. Pang, K. Xu, C. Du, N. Chen, J. Zhu, Improving adversarial robustness via promoting ensemble diversity, in: Int. Conf. Learn. Represent., in: Proceedings of Machine Learning Research, Vol. 97, PMLR, 2019, pp. 4970–4979.
[35] U. Shaham, Y. Yamada, S. Negahban, Understanding adversarial training: Increasing local stability of supervised models through robust optimization, Neurocomputing 307 (2018) 195–204.

[36] F. Tramèr, N. Papernot, I.J. Goodfellow, D. Boneh, P.D. McDaniel, The space of transferable adversarial examples, 2017, CoRR abs/1704.03453.
[37] M. Khoury, D. Hadfield-Menell, On the geometry of adversarial examples, 2018, CoRR abs/1811.00525.
[38] A. Fawzi, S. Moosavi-Dezfooli, P. Frossard, Robustness of classifiers: from adversarial to random noise, in: Adv. Neural Inform. Process. Syst., 2016, pp. 1624–1632.
[39] P. Tabacof, E. Valle, Exploring the space of adversarial images, in: International Joint Conference on Neural Networks, IJCNN, IEEE, 2016, pp. 426–433.
[40] A. Krizhevsky, Learning multiple layers of features from tiny images, 2009, URL https://api.semanticscholar.org/CorpusID:18268744.
[41] Y. Le, X. Yang, Tiny imagenet visual recognition challenge, CS 231N 7 (7) (2015) 3.
[42] S. Zagoruyko, N. Komodakis, Wide residual networks, in: R.C. Wilson, E.R. Hancock, W.A.P. Smith (Eds.), Brit. Mach. Vis. Conf., BMVA Press, 2016.
[43] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: IEEE Conf. Comput. Vis. Pattern Recog., IEEE Computer Society, 2016, pp. 770–778.
[44] T. Pang, M. Lin, X. Yang, J. Zhu, S. Yan, Robustness and accuracy could be reconcilable by (proper) definition, in: Int. Conf. Learn. Represent., 2022.
[45] K. Tong, C. Jiang, J. Gui, Y. Cao, Taxonomy driven fast adversarial training, in: AAAI, AAAI Press, 2024, pp. 5233–5242.
[46] Z. Wei, Y. Wang, Y. Guo, Y. Wang, CFA: Class-wise calibrated fair adversarial training, in: IEEE Conf. Comput. Vis. Pattern Recog., IEEE, 2023, pp. 8193–8201.
[47] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, in: Int. Conf. Learn. Represent., 2015.

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